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Computational methods in Physics

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Abstract

Lecture notes for the Computational methods in Physics class, within the Physical Sciences master degree of University of Pavia.

These notes are brief, and their purpose is to collect the main results and expressions, so that they can be used as a reference for an actual implementation.

With the notes are available a number of Mathematica notebooks, that serves as support material.

Contents

I Approximation and differentiation 3

1 Polynomial approximation 3

1.1 Nodal and Modal expansions . . . 3

1.1.1 Nodal expansion . . . 3

1.1.2 Modal expansion . . . 4

1.2 Barycentric forms . . . 5

1.3 Runge phenomenon . . . 6

1.4 Chebyshev polynomials . . . 7

1.5 Chebyshev barycentric weights . . . 9

1.6 Differentiation on the grid . . . 10

1.7 Integration on the grid . . . 12

2 Beyond single-grid construction 12

II Numerical Quadrature 12

III Integration of Differential Equations 13

Bibliography 14

Part I

Approximation and differentiation

1 Polynomial approximation

In the course, we will only cover the case of mono-dimensional (or univariate) problems. Let us simply mention that the multivariate case, with factorized domains (i.e. squares, cubes, etc) can be trivially obtained by combining multiple univariate constructions. If the domain is non-factorizable, then dedicated methods should be employed.

Suppose that we want to obtain a reliable approximation of a function $f(x)$, which might be very expensive to compute, so that we can generate the approximation once, and then use it instead of the function itself. This will be particularly useful for solving quadrature problems, and certain types of differential equations.

The point is how to construct such an approximation. We will take the general approach of using polynomials to approximate our solution. How to construct a polynomial in practice, we will see in a bit. For the moment, we state the Weierstrass approximation theorem, upon which relies the fact that we can expect a polynomial to perform sufficiently well as an approximant:

Weierstrass approximation theorem: If f is any continuous function on the finite closed interval $[a, b]$, then for every $\varepsilon > 0$ there exists a polynomial $p_n(x)$ of degree $n = n(\varepsilon)$ such that

$$\max_{x\in[a,b]} |f(x) - p_n(x)| < \varepsilon \tag{1}$$

□

There are two, conceptually distinct ways to go about finding a polynomial approximation of a given function. Both approaches rely on the definition of a polynomial basis that will define how the approximant is constructed. We will cover number of different basis, but the most famous are

$$\mathcal{M}_p = \{1, x, x^2, ..., x^p\} \tag{2}$$

which is nothing more than the natural basis for the polynomials of degree $\leq p$;

$$\mathcal{L}_p\left(\{x_i\}_{i=0}^p\right) = \{l_j(x; \{x_i\}_{i=0}^p)\}_{j=0}^p, \qquad l_j(x; \{x_i\}_{i=0}^p) = \prod_{\substack{i=0\\ i\neq j}}^{p} \frac{x - x_i}{x_j - x_k} \tag{3}$$

that is the Lagrange basis for the given set of nodes $\{x_i\}_{i=0}^p$, where note that the nodes are exactly $p + 1$, where p is the degree of each of the polynomials $l_j(x)$;

$$\{P_n(x)\}_{n=0}^p \tag{4}$$

which are the Legendre polynomials of degree $\leq p$. Note that the Legendre polynomials are just one example of a special class of polynomial families that are orthogonal with respect to some given inner product. In this case, the inner product looks like

$$\langle P_n|P_m\rangle = \int_{-1}^{1} dx P_n(x)P_m(x) = \frac{2}{2n+1}\delta_{n,m} \tag{5}$$

Let us note that neither the natural basis not the Lagrange basis are, in general, orthogonal basis.

Now, we can define the two different methods to find a polynomial approximation of a function $f(x)$ as nodal expansions versus modal expansions. The former is the easier to understand.

1.1 Nodal and Modal expansions

1.1.1 Nodal expansion

Fix a polynomial basis of degree p, with basis elements $b_j(x)$ then we can construct an approximation

$$\phi(x) = \sum_{j=0}^{p} c_j b_j(x) \quad \text{such that} \quad f(x_i) = \phi(x_i) \tag{6}$$

for a given set of exactly $p + 1$ distinct nodes x_i. Looking at the requirements, this should immediately point towards the Lagrange polynomials, and indeed they are essentially the polynomial basis of choice for nodal expansions. It

should also be apparent why the expansion is named nodal: we are fixing a set of nodes, on which we require that the approximation and the function itself are exactly equal. We can construct a bit of intuition why the Lagrange basis is the most obvious choice in this case. First of all, by the definition itself we see that

$$l_j(x_i) = \delta_{ij} \tag{7}$$

then, we have

$$\phi(x) = \sum_{j=0}^{p} c_j l_j(x) \Rightarrow \phi(x_i) = \sum_{j=0}^{p} c_j l_j(x_i) = \sum_{j=0}^{p} c_j \delta_{ij} = c_i \tag{8}$$

therefore, using the Lagrange basis, we immediately have that $c_i = f(x_i)$, which gives us a very straightforward way to construct the polynomial interpolation.

What would have happened if we were to choose a different basis? For instance, let us take the canonical polynomial basis:

$$\phi(x) = \sum_{j=0}^{p} c_j x^j \Rightarrow \phi(x_i) = \sum_{j=0}^{p} c_j x_i^j \tag{9}$$

Now, we can use a very common 'trick': let us collect $\phi(x_i)$ into a vector: $\boldsymbol{\phi} = \{\phi(x_0), ..., \phi(x_p)\}$, and let us do the same for the coefficients $\boldsymbol{c}$ and for the matrix $B_{ij} = x_i^j$. Then, the system of equations to determine the coefficients becomes

$$\boldsymbol{\phi} = B\boldsymbol{c} \Rightarrow \boldsymbol{c} = B^{-1}\boldsymbol{\phi} \tag{10}$$

Since we choose a basis for our polynomials, and we have the constant assumption that all the nodes are different from each other, then the matrix is guaranteed to be invertible (a zero eigenvalue in the matrix would signify a non-trivial set of nodes for which the linear independence of the basis functions would be violated, hence a contradiction). The problem is that, numerically, taking the inverse is both slow and potentially very inaccurate.

This procedure is not limited to the two set of polynomial basis, but can actually be employed for any polynomial basis, with the only modification being in the matrix B. We can also immediately see that the Lagrange basis is nothing more than the basis that diagonalizes the matrix B directly.

1.1.2 Modal expansion

The nodal expansion works in a rather general context, since the only required input is the choice of the polynomial basis and the selection of the nodes. The modal expansion instead requires the specification of the polynomial basis and an inner product on the space of polynomials. We saw that the Legendre polynomials are orthogonal with respect to the inner product with the unit function on the interval $[-1, 1]$.

As for the case of the nodal expansion, where the Lagrange polynomials played a special role, so here the entire class of orthogonal polynomials play a special role. Let us write again the polynomial approximation as

$$\phi(x) = \sum_{j=0}^{p} c_j b_j(x) \tag{11}$$

but now we want to derive the expansion coefficients by imposing the equality between the approximant and exact moments (or modes) of the function, i.e.

$$\langle f|b_j\rangle_\omega = \langle \phi|b_j\rangle_\omega \tag{12}$$

where ω defines the inner product as

$$\langle f|g\rangle_\omega = \int_{-1}^{1} dx\omega(x)f(x)g(x) \tag{13}$$

where we assume that the function $\omega(x) \geq 0$ for all $x \in [-1, 1]$. Note that here we will always assume that the domain is $[-1, 1]$. One can move to arbitrary compact domain via a simple linear transformation.

Here it becomes rather evident why orthogonal polynomials are the convenient choice: from

$$\langle b_i|b_j\rangle_\omega = n_i\delta_{i,j} \tag{14}$$

where n_i is some normalization, we immediately have that

$$\langle \phi|b_j\rangle_\omega = n_j c_j \Rightarrow c_j = \frac{1}{n_j}\langle f|b_j\rangle_\omega \tag{15}$$

We note that, typically, it is more computationally heavy determine the coefficient of a modal expansion compared to a nodal one. However, in some context it is very convenient being able to determine the expansion of an arbitrary

function onto a basis of polynomials. As an example, consider the following integro-differential problem that emerges in the study of perturbative-QCD

$$\frac{\partial f(x,\mu)}{\partial \log \mu^2} = \int_x^1 dz C(z) f\left(\frac{x}{z}\right) \tag{16}$$

where $f(x) \neq 0$ in $[0,1]$. Note that this problem cannot really be solved in this form. Then, assume that we are happy with an approximated solution, specifically, we are happy to have a polynomial expansion in the canonical basis. We could try to construct a nodal approximation (which is one of the current methods employed to solve such equations to a high degree of accuracy) or we could try to use a modal expansion. With the latter, we have

$$\begin{aligned}\frac{\partial \langle f(x,\mu)|x^n\rangle}{\partial \log \mu^2} &= \int_0^1 dx x^n \int_x^1 dz C(z) f\left(\frac{x}{z}\right) \\ &= \int_0^1 dz C(z) \int_0^z dx x^n f\left(\frac{x}{z}\right) \\ &= \int_0^1 dz C(z) z^{n+1} \int_0^1 dt t^n f(t)\end{aligned} \tag{17}$$

or, in other words

$$\frac{\partial \gamma_n(\mu)}{\partial \log \mu^2} = C_n \gamma_n(\mu) \tag{18}$$

where $\gamma_n(\mu) \equiv \langle f(x,\mu)|x^n\rangle$ and

$$C_n = \int_0^1 dz C(z) z^{n+1} \tag{19}$$

Now this is trivially solved as

$$\gamma_n(\mu) = \left(\frac{\mu}{\mu_0}\right)^{2C_n} \gamma_n(\mu_0) \tag{20}$$

This is especially powerful since, in practice, one may only be able to compute C_n instead of the full function $C(z)$, because the extra integral, simplifies the computation of the corresponding perturbative diagrams.

1.2 Barycentric forms

The naïve construction of the Lagrange interpolating polynomial

$$l_j(x) = \prod_{\substack{i=0\\ i\neq j}}^{p} \frac{x - x_i}{x_j - x_i} \tag{21}$$

is actually not very good in practice, both from a numerical stability point of view and from a performance point of view.

The first improvement that we can make to the formula is to note that we can precompute, once the grid is known, the weights

$$\lambda_j = \prod_{\substack{i=0\\ i\neq j}}^{p} (x_j - x_i)^{-1} \tag{22}$$

Moreover, we can define a unique polynomial of degree $p+1$ that has zeros on exactly the $p+1$ nodes of the grid

$$l(x) = \prod_{j=0}^{p} (x - x_j) \tag{23}$$

which is sometimes referred to as the node polynomial. Now, we can combine this two constructions to obtain a fairly simple alternative representation for the Lagrange polynomials

$$l_j(x) = \frac{\lambda_j l(x)}{x - x_j} \tag{24}$$

this is called the <u>first barycentric formula</u> of the Lagrange interpolant. What is the computational advantage here?

In the original formulation, to compute the approximant of a function

$$p(x) = \sum_{j=0}^{p} f_j l_j(x) \tag{25}$$

we needed to take $O(p)$ additions, and for each addition we needed $O(p)$ multiplications, therefore the total computational complexity comes out to be $O(p^2)$, which is typically not very good. With the first barycentric formula instead, we still need $O(p^2)$ operations to compute the weights λ_j, but this is a one-time-only computation, therefore it does not really matter that much. Once we have them, we do not have anymore nested sums and products: the product that defines $l(x)$ factorizes from the sum over j:

$$p(x) = l(x) \sum_{j=0}^{p} \frac{\lambda_j f_j}{x - x_j} \tag{26}$$

therefore the total computational complexity comes out to be $O(p + p) \sim O(p)$.

A useful observation is that we can obtain the weights as the derivative of the node polynomial evaluated on the grid nodes. Specifically

$$\lambda_j = \frac{1}{l'(x_j)} \tag{27}$$

The proof is by direct computation, and is left as a simple exercise.

There exist an even simpler formula though. We can notice that the polynomial approximant for the function $g(x) = 1$ is

$$\sum_{j=0}^{p} l_j(x) \tag{28}$$

but since the function itself is a polynomial, and we know that the polynomial interpolant is unique, we have immediately that

$$\sum_{j=0}^{p} l_j(x) = 1 \Rightarrow l(x) \sum_{j=0}^{p} \frac{\lambda_j}{x - x_j} = 1 \Rightarrow l(x) = \left[\sum_{j=0}^{p} \frac{\lambda_j}{x - x_j} \right]^{-1} \tag{29}$$

hence

$$p(x) = \left[\sum_{j=0}^{p} \frac{f_j \lambda_j}{x - x_j} \right] \left[\sum_{j=0}^{p} \frac{\lambda_j}{x - x_j} \right]^{-1} \tag{30}$$

which is called the <u>second barycentric formula</u> of the Lagrange interpolant.

There are proofs about the stability of the two formulas for different choices of the underlying grid. Specifically, for an arbitrary grid the first barycentric formula has stronger stability properties compared to both the naïve formulation and the second barycentric formula. It also behave quite well under extrapolation outside the interpolation domain. However, it suffers from potentially catastrophic finite-precision problems (under- and over-flows due to the finite precision of the representation of numbers on a computer). If one steers away from generic grids and restrict itself to special types of grid that have their points clustered near the edges of the domain (more on in in a moment), then the second barycentric formula works very well inside the interpolation region without suffering from finite-precision problems, whereas it is still not stable for the extrapolation region.

Conceptually, the main advantage of the second barycentric formula is that one can uniformly scale all the weights by some constant amount $\lambda_j \to c\lambda_j \ \forall j$ and the expression is unaffected. This is not true for the first barycentric formula, which has a linear dependence over the weights. It is simple to observe that the weights grow extremely fast (exponentially fast in fact) with the number of nodes in the grid. Therefore, once we have a lot of nodes, the weights are so big that cannot really be represented on a computer.

1.3 Runge phenomenon

To convince ourselves that using equispaced grids is almost never a good idea, we can try to interpolate the function

$$\frac{1}{1 + 25x^2} \tag{31}$$

on the standard interval $[-1, 1]$. If one does it on a grid of equispaced points, the more points one uses the worse the approximation of the function near the boundary of the intervals!

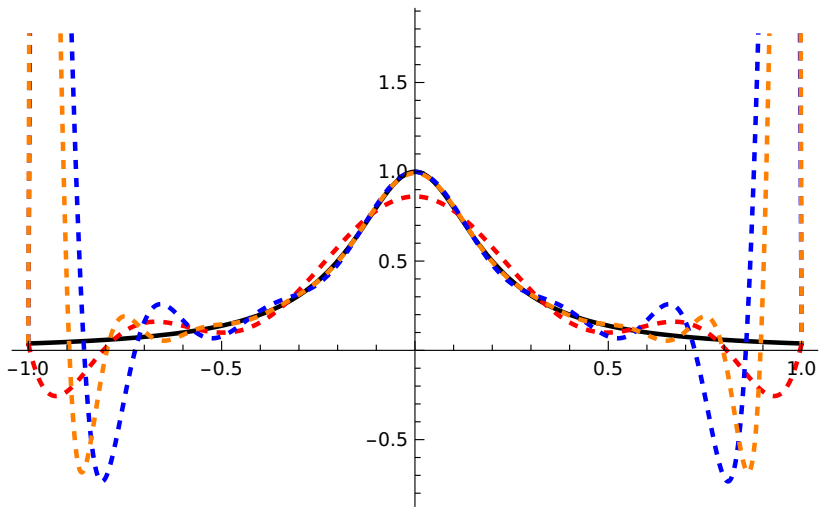

Figure 1: Exemplification of the Runge phenomenon. Solid black line is the true function, then in red, blue and orange respectively the interpolations with 10, 15, and 20 nodes.

1.4 Chebyshev polynomials

One of the central basis of orthogonal polynomials in approximation theory is the basis defined by the Chebyshev polynomials. These can be defined in different ways. The most compact representation, which seems a bit obscure at first glance, is

$$T_n(\cos(\theta)) = \cos(n\theta) \tag{32}$$

where we note that the support of the Chebyshev polynomials is always assumed to be the standard interval $[-1, 1]$. With this representation, it can be verified that they are orthogonal with respect to the scalar product

$$\begin{aligned} \langle T_n | T_m \rangle_{1/\sqrt{1-x^2}} &= \int_{-1}^{1} \frac{dx}{\sqrt{1-x^2}} T_n(x) T_m(x) \\ &= \int_0^{\pi} d\theta \cos(n\theta)\cos(m\theta) = \frac{\pi}{2}\delta_{nm}\left(1+\delta_{n0}\right) \end{aligned} \tag{33}$$

Although nice and compact, this representations is anything but clear. How can we be sure that T_n is in fact a polynomial? For once, we can immediately see that

$$T_0(\cos(\theta)) = 1, \quad T_1(\cos(\theta)) = \cos(\theta) \tag{34}$$

i.e.

$$T_0(x) = 1, \quad T_1(x) = x \tag{35}$$

Now, we can state the an inductive construction for the Chebyshev polynomials as

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x) \tag{36}$$

This can be easily verified to hold true for the cos representation above. And this recurrence relation clearly shows that T_n is a polynomial in its inputs for all n.

The more or less natural question is now: why the cosine? To understand it, assume to have a function f defined from our standard interval to the real numbers that is regular enough. Then, we can always write f as a modal expansion onto the basis of Chebyshev polynomials as

$$f(x) = \sum_{j\geq 0} a_j T_j(x) \tag{37}$$

Now consider a related function, that is defined by $F(\theta) = f(\cos(\theta))$, for which we obviously have that $F(\theta) = F(-\theta)$. For this function we can write the Fourier series as

$$F(\theta) = \sum_{k\in\mathbb{Z}} c_k e^{ik\theta} \tag{38}$$

where we have a series and not an integral because we are working on a compact interval and therefore have only numerably many distinct momentum modes, instead of a continuum spectrum. Since we know that the function is

even, we reduce the Fourier series in half, by noticing that the negative momentum modes must be identical to the positive ones, i.e.

$$F(\theta) = \sum_{k\geq 0} c_k \left(e^{ik\theta} + e^{-ik\theta}\right) = \sum_{k\geq 0} 2c_k \cos(k\theta) = \sum_{k\geq 0} 2c_k T_k(\cos(\theta)) \tag{39}$$

But we also have that

$$F(\theta) = f(\cos(\theta)) = \sum_{j\geq 0} a_j T_j(\cos(\theta)) \tag{40}$$

Since the expansion is done on a polynomial basis we automatically have that

$$2c_k = a_k \quad \forall k \tag{41}$$

In essence, the expansion on a Chebyshev basis is nothing more that a discrete cosine transform!

To make the connection complete, we need to specify what we mean by f being 'regular enough'. It is to consider f to be of uniform bound variation on the interval, which is to say that it should be Lipschitz-1, i.e. $\exists C$ such that $|f(x) - f(y)| \leq C|x - y|\ \forall x, y \in [-1, 1]$. In words, we can always find a finite constant C so that the variation of f on the interval is bounded by the separation between the points times the constant.

Then we can state the theorem
Chebyshev Series if $f : [-1, 1] \to \mathbb{R}$ is Lipschitz-1, then it has a unique Chebyshev series, which converges both absolutely and uniformly. $\square$

Leveraging again the connection between the Fourier representation and the Chebyshev one, we can introduce the Chebyshev-Lobatto points. To do so, fix an integer n, and consider the $n+1$ equispaced points on the top-half of the unit circle, so that $\theta_j = \pi(j/n)$ for $j = 0, .., n$. The cartesian coordinates of these points are clearly $(\cos(\theta_j), \sin(\theta_j))$. If we take the angles θ_j to be grid nodes for the Fourier representation, then we immediately have that grid nodes for the Chebyshev representation are the cosines of those angles, i.e.

$$t_j = \cos(\theta_j) = \cos\left(\frac{j\pi}{n}\right) \tag{42}$$

It is immediate to see that these are the unique $n+1$ points on which $T_n(x) = \pm 1$.

Why are we introducing these points? To answer we need to consider two different ways to construct an approximation of a function f using Chebyshev polynomials.

On the one side, we could use directly the modal expansion, truncated to some maximum degree, which is nothing more that the partial sum of the Chebyshev series. Let us define it as

$$f_p(x) = \sum_{j=0}^{p} a_j T_j(x) \tag{43}$$

On the other side, we want to construct a nodal interpolation, on the Chebyshev-Lobatto points. We know that the Lagrange polynomial is the unique interpolant. We also know that the Lagrange polynomials form a basis for the polynomials of a given degree, and so does the Chebyshev basis. Therefore, we can write the interpolant using the Chebyshev polynomials instead of the Lagrange ones as

$$p_n(x) = \sum_{j=0}^{p} c_j T_j(x) \tag{44}$$

Do not be tempted to interpret immediately $c_j = a_j$! c_j are defined by the interpolating construction, i.e.

$$p_n(t_k) = f(t_k) \tag{45}$$

which will give a system of linear equations in the c_j, whereas there is no guarantee that $f_p(t_k) = f(t_k)$!

However, intuition says that some connection must exist. Indeed, we can prove by direct computations that for fixed $p \geq 1$ and for any $m \in [0, p]$ the Chebyshev polynomials $T_{2jp\pm m}$ take the same values as T_m on the $p+1$ Chebyshev-Lobatto points for all $j > 0$.

With this, we can take the full Chebyshev of f, evaluate it in turn on each Chebyshev-Lobatto point as use the statement above, together with the nice convergence properties of the series, to reorder terms. Specifically, in the full series we re-arrange terms so that we have a tower of combinations of Chebyshev polynomials evaluated on the

grid points on which they all take the same values. This defines the coefficients c_k of the interpolant in terms of the coefficients of the full series as

$$c_0 = a_0 + \sum_{j\geq 1} a_{2jp} \tag{46}$$

$$c_k = a_k + \sum_{j\geq 1} a_{k+2jp} + a_{-k+2jp} \tag{47}$$

$$c_p = a_p + \sum_{j\geq 1} a_{p+2jp} \tag{48}$$

This, by construction, ensures that

$$p_n(x) = \sum_{j=0}^{p} c_j T_j(x) \tag{49}$$

interpolates the function on the grid.

This is obviously not a practical recipe to compute the interpolant! It is, however, very important from the theory point of view, because it ensures that the interpolants converge absolutely to the original function, which is a very useful property to have.

1.5 Chebyshev barycentric weights

Another major advantage of the Chebyshev-Lobatto points, is they allow for a closed-form computation of the interpolation weights, which immensely reduces the number of computations required in practice, not to mention the numerical stability of the procedure.

Let us start with some intermediate results. First, we have a representation for the Lagrange polynomial

$$l(x) = 2^{-p}\left(T_{p+1}(x) - T_{p-1}(x)\right) \tag{50}$$

To prove it, we know that $l(x)$ is defined by its $p+1$ roots, that for us are the $p+1$ Chebyshev-Lobatto points t_j. Since the polynomial on the right-hand side is also of degree $p+1$, if we can show that has the exact same roots, we have established the proportionality. Then, since the coefficient of x^{p+1} of $l(x)$ is exactly 1, we can just show that the same is true for the normalized right-hand side, and we are done.

First, let us compute

$$T_{p+1}(t_j) = \cos\left(\frac{(p+1)j\pi}{p}\right) = \cos(j\pi)\cos\left(\frac{j\pi}{p}\right) - \sin(j\pi)\sin\left(\frac{j\pi}{p}\right) \tag{51}$$

$$T_{p-1}(t_j) = \cos\left(\frac{(p-1)j\pi}{p}\right) = \cos(j\pi)\cos\left(\frac{j\pi}{p}\right) + \sin(j\pi)\sin\left(\frac{j\pi}{p}\right) \tag{52}$$

$$\tag{53}$$

Since $\sin(j\pi) = 0$, we immediately have that $T_{p+1}(t_j) - T_{p-1}(t_j) = 0$ for all $j \in [0, p]$. Now, from the defining recurrence relation of the Chebyshev polynomials we have that

$$T_n(x) = 2^{n-1}x^n + ... \tag{54}$$

if $n > 0$, from which immediately follows that

$$T_{p+1}(x) - T_{p-1}(x) = 2^p x^{p+1} + ... \tag{55}$$

from which the equality is established.

Now we can compute the weights. Recall that they can be defined from the derivatives of the Lagrange polynomial at the grid nodes, i.e.

$$\beta_j = \frac{1}{l'(t_j)} = \frac{2^p}{T'_{p+1}(t_j) - T'_{p-1}(t_j)} \tag{56}$$

where we switched to a dedicated notation for the weights, to remember that these are for the Chebyshev points only.

Now, we can compute the derivatives of the Chebyshev polynomials, using a very simple trick:

$$\frac{d}{dx}T_n(x) = \frac{d}{d\cos\theta}T_n(\cos\theta) = -\frac{1}{\sin\theta}\frac{d}{d\theta}T_n(\cos\theta) \Rightarrow \tag{57}$$

$$
\begin{aligned}
T'_{p\pm1}(t_j) &= -\frac{1}{\sin\theta_j}\frac{d}{d\theta_j}\cos\left((p\pm1)\theta_j\right) \\
&= -\frac{1}{\sin\theta_j}\frac{n}{\pi}\frac{d}{dj}\cos\left(j\pi\pm\frac{j\pi}{p}\right) \\
&= \frac{1}{\sin\theta_j}\frac{n}{\pi}\left(\pi\pm\frac{\pi}{p}\right)\left[\sin(j\pi)\cos\left(\frac{j\pi}{p}\right)\pm\cos(j\pi)\sin\left(\frac{j\pi}{p}\right)\right]
\end{aligned}
\tag{58}
$$

Now, if $j \neq 0, p$ we can trivially evaluate the expression, and get

$$
T'_{p\pm1}(t_j) = \pm(-1)^j(p\pm1) \tag{59}
$$

if $j = 0, p$, we need to be a bit more careful because the first one over sine diverges. No matter, taking the appropriate limit (by symmetry we have that the results for $j = p$ must be equal, up to a global $(-1)^p$ to the results of $j = 0$):

$$
T'_{p\pm1}(t_0) = (p\pm1)^2 \tag{60}
$$

Therefore, we have that

$$
\beta_0 = \frac{2^{p-1}}{p}\frac{1}{2} \tag{61}
$$

$$
\beta_{j\neq0,p} = \frac{2^{p-1}}{p}(-1)^j \tag{62}
$$

$$
\beta_p = \frac{2^{p-1}}{p}\frac{(-1)^p}{2} \tag{63}
$$

It is immediate to see that we can define a scaled down version of the weights by

$$
\tilde{\beta}_j = \frac{p}{2^{p-1}}\beta_j \tag{64}
$$

It is also convenient for the following discussions to extract the phase from the weights, and write

$$
\gamma_j = \tilde{\beta}_j(-1)^j \tag{65}
$$

having essentially that $\gamma_{0,p} = 1/2$ and $\gamma_{j\neq0,p} = 1$.

Therefore, our basis functions for the Chebyshev interpolant on the Chebyshev-Lobatto points is greatly simplified, especially in the second barycentric representation:

$$
p(x) = \sum_{j=0}^{p} f(t_j)b_j(x), \quad b_j(x) = \frac{\tilde{\beta}_j}{x-t_j}\Bigg/\sum_{i=0}^{p}\frac{\tilde{\beta}_j}{x-t_j} \tag{66}
$$

Now we can establish a connection between the coefficients $f(t_j)$ of the interpolant written in the natural Lagrange basis, and the coefficients c_k for the interpolant written in the Chebyshev basis. This will not be immediately useful, but we will need it when we will talk about quadrature rules. We can write

$$
p(x) = \sum_{j=0}^{p} f(t_j)b_j(x) = \sum_{j=0}^{p} c_j T_j(x) \tag{67}
$$

with

$$
c_j = \frac{2\gamma_j}{p}\sum_{i=0}^{p}\gamma_i T_j(t_i) f_i \tag{68}
$$

1.6 Differentiation on the grid

Now that we discussed the different ways to construct a polynomial interpolant, we can apply this construction to obtain a method to compute the derivatives of our function in an approximated way.

First, let us derive the results for the generic Lagrange interpolant, and then we will derive specialized representations for the Chebyshev grids.

First the trivial expression

$$
p(x) = \sum_{j=0}^{p} f_j l_j(x) \Rightarrow p'(x) = \sum_{j=0}^{p} f_j l'_j(x) \tag{69}
$$

but then $p'(x)$, as a function, can be itself interpolated on the grid, to give the representation

$$p'(x) = \sum_{j=0}^{p} p'(x_j) l_j(x) \tag{70}$$

We can use now the first representation into the second, to obtain

$$p'(x) = \sum_{j=0}^{p} l_j(x) \sum_{i=0}^{p} f_i l_i'(x_j) \tag{71}$$

Define the derivative matrix

$$D_{ij} = l_i'(x_j) \tag{72}$$

and we can write

$$f'(x) \sim p'(x) = \sum_{i=0}^{p} f_i \sum_{j=0}^{p} D_{ij} l_j(x) \tag{73}$$

i.e. we have identified

$$l_i'(x) = \sum_{j=0}^{p} D_{ij} l_j(x) \tag{74}$$

which also gives us a formula for repeated differentiation: with the convention that repeated indexes are summed over

$$\frac{d^n}{dx^n} l_{i_0}(x) = D_{i_0 i_1} D_{i_1 i_2} .. D_{i_{n-1} i_n} l_{i_n}(x) \tag{75}$$

Now we can explicitly compute the derivative matrix. First, we note that we have

$$\begin{aligned} l_j'(x) &= \frac{d}{dx} \prod_{\substack{k=0 \\ k \neq j}}^{p} \frac{x - x_k}{x_j - x_k} \\ &= \sum_{\substack{n=0 \\ n \neq j}}^{p} \frac{1}{x_j - x_n} \prod_{\substack{k=0 \\ k \neq j,n}}^{p} \frac{x - x_k}{x_j - x_k} \\ &= \left[\sum_{\substack{n=0 \\ n \neq j}}^{p} \frac{1}{x - x_n} \right] l_j(x) \end{aligned} \tag{76}$$

Now we can use this expression to evaluate it at the grid points. First, we have the diagonal terms

$$l_j'(x_j) = \left[\sum_{\substack{n=0 \\ n \neq j}}^{p} \frac{1}{x_j - x_n} \right] l_j(x_j) = \sum_{\substack{n=0 \\ n \neq j}}^{p} \frac{1}{x_j - x_n} \tag{77}$$

For the off-diagonal terms we instead obtain

$$l_j'(x_i) = \left[\sum_{\substack{n=0 \\ n \neq j}}^{p} \frac{1}{x_j - x_n} \right] \prod_{\substack{k=0 \\ k \neq j,n}}^{p} \frac{x_i - x_k}{x_j - x_k} = \frac{1}{x_j - x_i} \prod_{\substack{k=0 \\ k \neq j,i}}^{p} \frac{x_i - x_k}{x_j - x_k} \tag{78}$$

where the last equality holds because if $\forall n \neq i$ in the sum, the product contains a single term in the numerator that is $x_i - x_i$, hence it removes the corresponding contribution.

Notice that we obtained the results in the naïve representation for the Lagrange interpolant. It is actually easy to see that an equivalent presentation of the final result is

$$f'(x) \sim p'(x) = \sum_{j=0}^{p} l_j(x) \sum_{i=0}^{p} f_i D_{ij} \tag{79}$$

with this we can define the coefficients $\tilde{f}_j = \sum_{i=0}^{p} f_i D_{ij}$, and with these apply the first or second barycentric formulas. For instance

$$p'(x) = l(x) \sum_{j=0}^{p} \frac{\tilde{f}_j \lambda_j}{x - x_j} \tag{80}$$

and equivalently for the second barycentric formula.

For the Chebyshev-Lobatto grids, we can derive a closed form representation for the derivative matrix. We already saw the derivative of the Chebyshev polynomials in the previous section, but here can can make use of a nice closed form representation for it, which is more convenient for this derivation:

$$T_k'(x) = kU_{k-1}(x) \tag{81}$$

where U_k are the Chebyshev polynomials of the second kind, which can defined via

$$U_k(\cos(\theta)) = \frac{\sin((k+1)\theta)}{\sin(\theta)} \tag{82}$$

They also from a set of orthogonal polynomials, with respect to the metric $\sqrt{1-x^2}$, i.e.

$$\int_{-1}^{1} dx\sqrt{1-x^2}\; U_n(x)U_m(x) \propto \delta_{nm} \tag{83}$$

To derive now the derivative matrix we can rely on the Chebyshev interpolant written as at the end of the last section:

$$p(x) = \sum_{j=0}^{p} c_j T_j(x), \qquad c_j = \frac{2\gamma_j}{p}\sum_{i=0}^{p}\gamma_i T_i(t_j) f_j \tag{84}$$

from which we have

$$p'(x) = \sum_{j=0}^{p} c_j T_j'(x) = \sum_{j=0}^{p}\left[\frac{2\gamma_j}{p}\sum_{i=0}^{p}\gamma_i T_j(t_i) f_i\right] jU_{j-1}(x) \tag{85}$$

Now we know from the generic discussion that if we evaluate $p'(x)$ on the grid node x_j and we look at the coefficient of the i-th value f_i this gives us the entry D_{ij}. So, rearranging a bit the expression above, we reach the following

$$p'(t_j) = \sum_{i=0}^{p} f_i\left[\sum_{k=0}^{p}\frac{2\gamma_k}{p} kT_k(t_i)U_{k-1}(t_j)\gamma_i\right] \tag{86}$$

To take the sums, one can write all the sine and cosine functions in exponential form. A bit of care is needed for the diagonal terms, and for when either of the i, j indexes is $0, p$. No matter, with patience and the help of Mathematica, one can arrive at the following representation

$$D_{00} = -D_{pp} = \frac{2p^2+1}{6} \tag{87}$$

$$D_{ii} = -\frac{t_i}{2(1-t_i^2)} \tag{88}$$

$$D_{ij} = -\frac{\gamma_i}{\gamma_j}\frac{(-1)^{i+j}}{t_i - t_j} \tag{89}$$

This part has been adapted from the discussion given in [1], with the notable exception that the derivative matrix presented here is the transpose of the derivative matrix of the paper, and notation is slightly different.

1.7 Integration on the grid

2 Beyond single-grid construction

Part II

Numerical Quadrature

Part III

Integration of Differential Equations

Bibliography

[1] Markus Diehl, Riccardo Nagar, and Frank J. Tackmann. "ChiliPDF: Chebyshev interpolation for parton distributions". In: Eur. Phys. J. C 82.3 (2022), p. 257. DOI: 10.1140/epjc/s10052-022-10223-1. arXiv: 2112.09703 [hep-ph].